

# New lower bounds for the degree/diameter problem via interaction with a browser-accessible LLM

Ryosuke Mizuno  
A.I.System Research Co., Ltd.  
`mizuno@aisr.dev`

5 June 2026

## Abstract

Let  $N(\Delta, D)$  denote the maximum number of vertices in a simple undirected connected graph with maximum degree at most  $\Delta$  and diameter at most  $D$ . Determining  $N(\Delta, D)$  is known as the degree/diameter problem. Although the problem has been studied for many years, the exact value of  $N(\Delta, D)$  is unknown for most pairs  $(\Delta, D)$ . In this paper we construct explicit graphs, using a route-chart lift construction discovered through interaction with ChatGPT via its standard web interface, showing that

$$N(12, 5) \geq 34,992, \quad N(16, 5) \geq 147,456.$$

These improve the corresponding recorded lower bounds 29,621 and 132,496. The search was conducted through ordinary dialogue in the standard ChatGPT web interface, without a dedicated agentic system, external orchestration, problem-specific search engine, or formal proof assistant set up in advance by the author. As far as the visible transcript shows, the author did not prompt the model with the concrete components of the construction, nor with a problem-specific proof or construction strategy.

After presenting the mathematical result, we describe the discovery process on the basis of the visible transcript. We focus on the general, problem-independent prompts and meta-level interventions under which the approximately six-day search proceeded, and we identify the stage at which the abstraction underlying the construction first appeared.

**MSC 2020:** 05C12, 05C25, 05C76.

**Keywords:** degree/diameter problem, graph theory, LLM-assisted mathematical discovery, mathematical reasoning, human–AI collaboration, ChatGPT, transcript analysis.

---

## 1. Introduction

Recent work has rapidly expanded the use of large language models (LLMs), neural language models, and systems built around them for mathematical construction search, theorem proving, problem generation, and formalization support. Representative peer-reviewed examples include FunSearch [RomeraParedes2024], which obtains mathematical constructions by combining a pretrained LLM

with a systematic evaluator in a program-search loop, and AlphaGeometry [Trinh2024], which combines a neural language model with a symbolic deduction engine for olympiad geometry. Other examples include AlphaProof [Hubert2026], a reinforcement-learning approach to formal proof search using Lean as a verifiable environment, and TongGeometry [Zhang2026], which addresses the proposal and proof of olympiad geometry problems by guided tree search and a fine-tuned LLM.

From late 2025 through May 2026, preprints, technical reports, and public reports from research institutions have also described the use of AI/LLMs in mathematical discovery. These reports include examples of counterexample discovery, lower-bound improvement, construction search, and formal-verification support [Bubeck2025, Woodruff2026, NagdaRaghavanThakurta2026, BhanNobiliLanger2026, JuEtAl2026, PrzybockiMackeyHeuleSubercaseaux2026, TsoukalasEtAl2026, OpenAI2026, AlonEtAl2026]. Many of these examples rely on dedicated evaluators, evolutionary search, tree search, formalizers, symbolic reasoning engines, theorem retrieval, SAT solvers, the Lean prover, or problem-specific evaluation loops. The present paper considers a different and more minimal setting:

*How far can mathematical search and discovery proceed using only an LLM available through a standard browser interface?*

This question is practically relevant not only for laboratories able to build large dedicated systems, but also for individual researchers and developers who use LLMs for mathematical exploration. Even when a dedicated agentic system or orchestration is eventually built, it is useful to understand what kind of control is possible over a readily available LLM, and how far that baseline can reach. As a case study in this setting, this paper documents a new lower-bound construction for the degree/diameter problem. The construction was obtained through a long dialogue with ChatGPT in its standard web interface, under general, problem-independent prompts and meta-level interventions.

The degree/diameter problem asks for the maximum number of vertices in a simple undirected graph with bounded maximum degree and bounded diameter. In the notation used here, the maximum degree is at most  $\Delta$ , so each vertex has at most  $\Delta$  neighbors, and the diameter is at most  $D$ , so any two vertices are connected by a path of length at most  $D$ .

Intuitively, when each vertex has at most  $\Delta$  incident edges, the number of vertices reachable within distance  $D$  from a fixed vertex is maximized in the tree-like case, where no branches meet up to and including depth  $D$ . This gives the Moore upper bound

$$N(\Delta, D) \leq 1 + \Delta \sum_{i=0}^{D-1} (\Delta - 1)^i = \frac{\Delta(\Delta - 1)^D - 2}{\Delta - 2} \quad (\Delta \geq 3).$$

For many pairs  $(\Delta, D)$  there is no Moore graph attaining this bound, and the construction of large explicit examples has long been studied [HoffmanSingleton1960, Damerell1973, Bannai1973, MillerSiran2013]. Known constructions include Cayley graphs, voltage graphs, compound graphs, and other computational and algebraic constructions, which have played an important role in setting records for fixed parameters [BermondDelormeFarhi1984, BermondDelormeQuisquater1986, LozSiran2008, Comellas2026].

The mathematical contribution of this paper is a construction, obtained by integrating existing tools for the degree/diameter problem, that establishes the new lower bounds

$$N(12, 5) \geq 34,992, \quad N(16, 5) \geq 147,456.$$

The main text first states the theorem, compares it with the known lower bounds, and outlines the construction principle. Precise definitions, proofs, and verification procedures are in Appendix A. We then describe how the construction principle emerged during the dialogue, and how its appearance was temporally related to failures, changes of direction, and meta-level interventions by the author.

This paper does not treat ChatGPT as an author or as an autonomous prover. ChatGPT was used as an interactive discovery engine. The author checked the mathematical claims, code, certificates, citations, and final manuscript, and takes responsibility for the contents of the paper.

The LLM-generated files included in the supplementary materials are, in principle, preserved in the form obtained during the dialogue, in order to record the provenance of the discovery process and of the generated construction. The mathematical claims of the paper are supported by the explicit construction, the finite list of certificate conditions, and the verification procedures that check them, all of which are given in the main text and in Appendix A.

## 1.1 Contributions of this paper

This paper makes two contributions.

1. **Improved mathematical lower bounds.** For the degree/diameter problem with diameter 5, we give explicit simple undirected graph constructions establishing the new lower bounds  $N(12, 5) \geq 34,992$  and  $N(16, 5) \geq 147,456$ .
2. **A record and analysis of a discovery process involving a browser-accessible LLM.** We describe a long dialogue with ChatGPT in its standard web interface, including unsuccessful search intervals. The dialogue was conducted without a dedicated agentic system, custom orchestration, systematic evaluator, problem-specific search engine, or formal proof assistant set up in advance by the author. We also record, as process tracing rather than as a causal claim, that the author’s problem-independent meta-level interventions occurred near several junctures at which the search moved toward abstraction and finite certificate design. These interventions adjusted the constraints and acceptance criteria of the search.

These two contributions can be read independently. The mathematical result in Contribution 1 can be verified from the explicit construction and the finite certificates in Appendix A alone. Its correctness does not depend on the fact that the construction was discovered through dialogue with a browser-accessible LLM. Contribution 2 is an observational report of a single case concerning the process that led to the discovery, and its outline can be understood without referring to Appendix A.

---

## 2. The degree/diameter problem and the main result

### 2.1 Basic graph-theoretic terminology and the degree/diameter problem

The graphs considered in this paper consist of finitely many vertices and edges joining pairs of vertices. A graph whose edges are oriented is called a **directed graph**, and a graph whose edges are not oriented is called an **undirected graph**. An undirected graph with no loops and no multiple edges is called a **simple undirected graph**. A graph is **connected** if every vertex can be reached from every other vertex by a path along edges.

The number of edges incident to a vertex  $v$  is the **degree** of  $v$ . Saying that the maximum degree of a graph is at most  $\Delta$  means that every vertex has at most  $\Delta$  neighbors. A graph in which all vertices have the same degree is called **regular**. The **distance** between two vertices  $u, v$  is the minimum length of a path from  $u$  to  $v$ . The **diameter** of a graph is the maximum distance over all pairs of vertices.

For integers  $\Delta, D \geq 1$ , define

$$N(\Delta, D)$$

to be the maximum number of vertices in a simple undirected connected graph with maximum degree at most  $\Delta$  and diameter at most  $D$ . The degree/diameter problem is the problem of determining  $N(\Delta, D)$ . In the diameter-5 case treated here, the goal is to make the number of vertices as large as possible under the constraint that any two vertices are joined by a path of length at most 5. For example, constructing a 12-regular graph of diameter 5 means explicitly giving a large graph in which every vertex has degree 12 and every pair of vertices is joined within 5 steps.

## 2.2 Known lower bounds and the improvement in this paper

Comellas’s Mendeley Data V11 collects the largest known  $(\Delta, D)$ -graphs as of January 2026 [Comellas2026, ComellasData2026]. In the related web table, checked on 31 May 2026, the entries relevant to this paper had the same lower-bound values. The known lower bounds for diameter 5 relevant to this paper are as follows.

| $(\Delta, D)$ | Known lower bound | Source                 |
|---------------|-------------------|------------------------|
| (12, 5)       | 29,621            | [Comellas2026]         |
| (16, 5)       | 132,496           | [DelormeBipartite1985] |

This paper improves these to 34,992 and 147,456, respectively.

**Theorem 2.1.** There exist explicit simple undirected graphs  $G_3, G_4$  satisfying the following.

1.  $G_3$  is 12-regular,  $|V(G_3)| = 34,992$ , and  $\text{diam}(G_3) = 5$ .
2.  $G_4$  is 16-regular,  $|V(G_4)| = 147,456$ , and  $\text{diam}(G_4) = 5$ .

Theorem 2.1 immediately implies

$$N(12, 5) \geq 34,992, \quad N(16, 5) \geq 147,456.$$

The improvements are summarized below.

| Parameter | Previous LB | This paper | Improvement       | Moore UB | Moore ratio |
|-----------|-------------|------------|-------------------|----------|-------------|
| (12, 5)   | 29,621      | 34,992     | +5,371 (+18.13%)  | 193,261  | 18.11%      |
| (16, 5)   | 132,496     | 147,456    | +14,960 (+11.29%) | 867,857  | 16.99%      |

Here the Moore ratio is the lower bound obtained in this paper divided by the corresponding Moore upper bound.

## 2.3 Outline of the construction: controller and fiber

The precise construction and proofs are given in Appendix A. This section gives a guide to the construction principle.

A vertex of the final graph has the form

$$(c, x) \in V(C) \times \mathbb{F}_q^5,$$

where  $\mathbb{F}_q$  is a finite field,  $c$  is a vertex of a small auxiliary graph  $C$ , and  $x$  is a coordinate in the copy of  $\mathbb{F}_q^5$  lying above  $c$ . This copy,

$$\{c\} \times \mathbb{F}_q^5,$$

is called the **fiber** over  $c$ .

We call  $C$  the **controller**. It determines the coarse routing of the final graph. Edges are added to the vertex set  $V(C) \times \mathbb{F}_q^5$  as follows.

1. Choose a  $\rho$ -regular simple undirected graph  $C$ . Let  $S$  be a finite set with  $|S| = \rho$ , equipped with a fixed-point-free involution  $t \mapsto t^{-1}$ , so that

$$(t^{-1})^{-1} = t, \quad t^{-1} \neq t.$$

Such an  $S$  is called a symmetric alphabet.

Regard each undirected edge of  $C$  as two directed edges, and assign an element of  $S$  to each directed edge. We require that, at each vertex  $u \in V(C)$ , the  $\rho$  directed edges leaving  $u$  receive the elements of  $S$  exactly once, and that if a directed edge  $u \rightarrow v$  is assigned  $t \in S$ , then the reverse directed edge  $v \rightarrow u$  is assigned  $t^{-1}$ . Such a labeling is called an **inverse-consistent labeling**.

2. To each label  $t \in S$  assign

$$A_t \in \text{GL}_5(\mathbb{F}_q), \quad b_t \in \mathbb{F}_q^5,$$

with the inverse-symbol conditions

$$A_{t^{-1}} = A_t^{-1}, \quad b_{t^{-1}} = A_t^{-1}b_t.$$

This assignment is called a **route chart**. It specifies how fiber coordinates in  $\mathbb{F}_q^5$  move along the edges of  $C$ .

3. Let  $t \in S$  be the label assigned to a directed edge  $u \rightarrow v$  of  $C$ . For  $(u, x), (v, y) \in V(C) \times \mathbb{F}_q^5$ , join  $(u, x)$  and  $(v, y)$  by an edge if there exists  $\lambda \in \mathbb{F}_q$  such that

$$y = A_t x + \lambda b_t.$$

The graph obtained by lifting the edges of  $C$  to  $V(C) \times \mathbb{F}_q^5$  by this rule is called a **route-chart lift graph**.

In a route-chart lift, the following two additional conditions guarantee diameter at most 5.

1. **Controller-side condition.** For any controller vertices  $c, d \in V(C)$ , there is a nonbacktracking walk of length exactly 5 from  $c$  to  $d$ . Here nonbacktracking means that the walk never immediately reverses the edge just traversed. A controller with this property is called an **exact-NB-5 controller**.

2. **Fiber-side condition.** A sequence of elements of  $S$  is called a **word**. A word  $w = t_1 t_2 t_3 t_4 t_5$  of length 5 is **reduced** if

$$t_{i+1} \neq t_i^{-1} \quad (1 \leq i < 5).$$

The fiber-side condition is that, for every reduced word  $w = t_1 t_2 t_3 t_4 t_5$  of length 5, the controllability matrix

$$M_w = [A_{t_5} \cdots A_{t_2} b_{t_1} \mid A_{t_5} \cdots A_{t_3} b_{t_2} \mid A_{t_5} A_{t_4} b_{t_3} \mid A_{t_5} b_{t_4} \mid b_{t_5}] \in M_5(\mathbb{F}_q)$$

is nonsingular.

If the controller and fiber data satisfy these conditions, the following proposition shows that the resulting graph has diameter at most 5.

**Proposition 2.2 (route-chart lift principle).** Let  $C$  be a  $\rho$ -regular simple undirected graph and let  $S$  be a symmetric alphabet with  $|S| = \rho$ . Suppose that the directed edges of  $C$  are given an inverse-consistent labeling. Suppose also that a universal route chart on  $\mathbb{F}_q^s$  is given, meaning that the controllability matrix is nonsingular for every reduced word of length  $s$ . If  $C$  is an exact-NB- $s$  controller, then this data defines a simple  $\rho q$ -regular graph  $G$  on  $V(C) \times \mathbb{F}_q^s$ , and

$$|V(G)| = |V(C)|q^s, \quad \text{diam}(G) \leq s.$$

The proof is given in Appendix A. The key point is that the controller-side condition and the fiber-side condition are independent. The controller-side condition is a property of the controller alone, independent of the fiber. The fiber-side condition depends only on the route chart over the symbol set  $S$ , independent of the controller.

Thus the problem of constructing a large graph with fixed degree and diameter is reduced to two independent finite design problems.

| Component   | Role                                                                   | What to verify                                                                 |
|-------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| controller  | a small graph determining the coarse routing of the final graph        | every pair of controller vertices is joined by a length-5 nonbacktracking walk |
| fiber chart | linear-algebraic data moving fiber coordinates along controller routes | $M_w$ is nonsingular for every reduced word of length 5                        |

In the construction of this paper, the controller, the fiber chart, and the final graph are built as follows.

First, from the 6-vertex 4-regular loopless multigraph  $B$  in Appendix A.5, together with a voltage assignment with values in

$$H = C_4 \times C_6,$$

we obtain a 144-vertex simple 4-regular connected graph  $C$  by the standard voltage lift defined in Appendix A.2. For this  $C$ , finite computation verifies that every ordered pair of vertices  $(c, d)$  is joined by a nonbacktracking walk of length exactly 5. This graph  $C$  is the controller.

Next, we give the directed edges of  $C$  an inverse-consistent labeling by

$$S = \{a, a^{-1}, b, b^{-1}\}.$$

We also give explicit route charts on  $\mathbb{F}_3^5$  and  $\mathbb{F}_4^5$ , and verify by finite computation that the controllability matrix is nonsingular for every reduced word of length 5.

Applying Proposition 2.2 with  $q = 3$  gives a 12-regular graph on

$$144 \cdot 3^5 = 34,992$$

vertices. Applying it with  $q = 4$  gives a 16-regular graph on

$$144 \cdot 4^5 = 147,456$$

vertices. In both cases the diameter is at most 5.

Finally, each of these vertex counts exceeds the corresponding Moore upper bound for diameter 4. Hence each of the two obtained graphs has diameter exactly 5, and Theorem 2.1 follows.

The construction is closely related to the existing theory of graph lifts and voltage graphs. The controller in §A.5 is an ordinary group-valued voltage lift. The final route-chart lift can also be described as a lift by a permutation voltage assignment after each directed edge of the controller is expanded into  $q$  directed edges indexed by  $\lambda \in \mathbb{F}_q$ . The relation between the present construction and existing methods is discussed in §7.

---

### 3. Dialogue environment

The search was conducted by the author through ordinary dialogue in the standard ChatGPT web interface. The author did not use proprietary external orchestration, a custom agent framework, a problem-specific search engine prepared in advance, or a formal proof assistant. The setup was as follows.

- User interface: ChatGPT in a web browser.
- Model display name: `gpt-5-5-pro` in the transcript metadata.

In this paper, “browser-accessible” means that the external workflow built and used by the author was limited to the standard ChatGPT web interface. The internal implementation of the LLM, its inference process, and any context-management mechanisms that may have been used on the service side are not observable; the paper therefore makes no claim that these mechanisms were simple.

All dialogue was conducted in Japanese. The user prompts quoted in §4–§5 are the author’s English translations; the original Japanese messages are preserved in the supplementary visible transcript (Appendix B).

Because the ChatGPT account appeared to have access to information from other sessions in the same account, this paper does not claim that the main transcript released here was, by itself, the sole source of information for the search. To the extent verified by the author, the situation was as follows.

- Before Round 038 of the main session released here, other sessions contained exchanges on the degree/diameter problem for  $(\Delta, D) = (8, 2), (6, 4)$  and on the degree-57 Moore graph, as well as exchanges on other mathematical tasks.

- However, no other session before Round 038 was confirmed to contain a general degree/diameter construction or the route-chart lift method ultimately used in this paper.
- After Round 038, a separate session was started to verify the first lower-bound-improvement candidate obtained in Round 036.
- After Round 042, in which the creation of handoff materials for other sessions was requested, two separate sessions in the same account began, in parallel, a search for generalizations of the result.
- Neither separate session reached either the ideas that appeared in Rounds 049–050 of the released main session or the final lower bounds shown in §2. The released main session was the first to reach the present result.

---

## 4. The discovery process

The observations below concern Rounds 001–052 of the ChatGPT sessions, conducted from 22 to 27 May 2026. These are the rounds relevant to the present discovery and verification. They include the initial literature survey, construction attempts, failure reports, changes of direction, code generation, and verification results. Unrelated exchanges from Round 053 onward are excluded from the analysis and from the supplementary transcript.

The dialogue log was extracted from the conversation log downloaded through the official ChatGPT export function. The supplementary script `dump_ddp_visible.py` restricts the extraction to browser-visible user prompts and assistant answers. In this paper, one user prompt together with one or more responses to it is organized as a round of the transcript. The full visible transcript of Rounds 001–052 is included in the supplementary materials. The structure of the supplementary materials is summarized in Appendix B.

### 4.1 The initial prompt

The dialogue began in Round 001 with the author giving a high-level goal of roughly the following form.

- Aim to set a new record for the degree/diameter problem within  $\Delta \leq 16$ ,  $D \leq 10$ .
- First, survey the relevant literature.
- Then attempt a construction based on an idea not essentially contained in the literature.
- Avoid mere combinations of existing methods, parameter increases, substitutions of the algebraic system, and non-novel local search around existing solutions.
- During the idea-generation and construction phases, refrain from web search in principle.
- The LLM may choose  $\Delta, D$  based on the suitability of its idea.

Except for specifying the target range of degree and diameter, the initial prompt consisted of fairly general research instructions. It asked the model to first conduct a literature survey, to attempt a proof using an idea not essentially contained in the literature, and to suppress web search during the ideation and construction stages. The examples in the fourth item are specific to the degree/diameter problem, but they were intended only to clarify what was meant by “not essentially contained” in the literature.

Within the visible transcript analyzed in this paper, the author did not directly specify the concrete components or candidate families of the final construction. Instead, the author provided general

constraints and acceptance criteria, including differentiation from the literature, suppression of web search, withdrawal from local repair, stopping and reorganization, and requests to explain prospects of success in advance. Candidate generation itself was left mainly to the LLM.

## 4.2 Initial exploration

In the initial exploration from Round 001 to Round 021, many ideas were examined, including what the transcript labeled as non-regular permutation lifts, thinning of finite geometries, matching contraction, Hashimoto / de Bruijn type constructions, rank-controller type constructions, random regular graphs, and affine Schreier cover type constructions. Most of these did not ultimately lead to a certificate.

## 4.3 The first essential transition: route as coordinate chart

The first substantive progress occurred in Round 023, when the principle of the controllable route-chart lift appeared. This principle is the core of the final construction outlined in §2.3. In Rounds 022 and 023, the user requested “an a priori mathematical reason to expect success” and asked the LLM “to devise a new principle.” In Round 023, the LLM presented the following view.

*A base/controller walk is not merely a displacement. A controller route of length  $k$  should be a coordinate chart that covers all of  $\mathbb{F}_q^k$  through the  $q^k$  port choices of the fiber.*

To explain this abstraction, we first recall the standard group-valued voltage-lift point of view. In an ordinary voltage lift, one attaches a group element to each edge of the base graph. Choosing a single path on the base graph from a start vertex to an end vertex determines a single product or sum of group elements along that path, and hence a single displacement in the fiber. Thus, to reach many fiber targets, many base paths must produce sufficiently different displacements. Even if the total number of short paths is large, some targets remain unreachable if their displacements coincide too often.

The route-chart lift point of view is different. Even after fixing a single route on the controller, one may choose  $\lambda_i \in \mathbb{F}_q$  at each step. For a route of length 5, the available control sequences are

$$(\lambda_1, \dots, \lambda_5) \in \mathbb{F}_q^5,$$

of which there are  $q^5$ . If the map from these sequences to the final fiber displacement is a linear isomorphism, then a single controller route is not merely a single move. It parametrizes all target points in the fiber.

This idea decomposed the search for the huge final graph into two finite design problems.

1. **Controller problem.** Construct a small 4-regular graph that joins any pair of controller vertices by a nonbacktracking walk of length exactly 5.
2. **Fiber chart problem.** Construct a chart over a finite field such that, for every reduced word of length 5, the corresponding  $5 \times 5$  controllability matrix is nonsingular.

This decomposition is central to the discovery process. Rather than directly guessing the whole final graph, the LLM reformulated the task as the separate construction of a small controller-side graph design and a finite list of fiber-side linear-algebraic conditions, followed by their composition through a linear-algebra lemma.

#### 4.4 Success of the fiber chart and the controller bottleneck

In the proof of concept in Round 024, affine control data were found over  $\mathbb{F}_4^4$  such that the controllability matrix associated with every reduced word of length 4 had full rank. This confirmed that the fiber-chart principle could hold.

Constructing the corresponding controller was harder. For example, targeting (16, 4) requires a 4-regular controller with at least 58 vertices in which all ordered pairs are joined by a nonbacktracking walk of length exactly 4. In Rounds 026–033, many cyclic voltage lifts, direct searches, and near-controller repairs were tried, but none reached a certificate.

The important observation at this stage was the identification of a bottleneck: for the shortest (16, 4) target, the controller-side constraints were too tight.

#### 4.5 Transition to diameter 5 and the first complete candidate

In response to this bottleneck, the search moved to  $s = 5$  from Round 034 onward. In Rounds 034–035, the examination of controller candidates continued with the length extended. Increasing the length to 5 gave more room for controller design, and in Round 036 the first complete candidate was obtained by combining a universal chart on  $\mathbb{F}_4^5$  with a 132-vertex exact-NB-5 controller:

$$\Delta = 16, \quad D \leq 5, \quad n = 132 \cdot 4^5 = 135,168.$$

This was the first candidate graph produced from the route-chart principle to improve a degree/diameter lower bound.

#### 4.6 Post-completion generalization search and the update to the final controller

In Rounds 037–052, after the first complete candidate had been obtained, a generalization search was carried out, and the 132-vertex controller from Round 036 was replaced by the final 144-vertex controller.

First, in Round 039, combining the same 132-vertex controller with the  $\mathbb{F}_3^5$  chart yielded a (12, 5) candidate on 32,076 vertices. In Rounds 040–048, charts over  $\mathbb{F}_5^5$ ,  $\mathbb{F}_5^4$ , and  $\mathbb{F}_2^5$ , cubic controllers, and extensions to diameters 4 and 6 were also attempted. In many of these attempts, the chart-side condition was achieved, but no corresponding controller was found.

In Round 049, the user requested a focus on “ideas for which there is an a priori mathematical reason to expect that they have the property currently desired.” The LLM narrowed the search to a controller strategy based on  $C_4 \times C_6$ . In the subsequent Round 050, a 144-vertex exact-NB-5 controller was obtained from a 6-vertex 4-regular multibase and  $C_4 \times C_6$  voltage coverage. Composing it with the  $\mathbb{F}_4^5$  chart yielded

$$N(16, 5) \geq 147,456.$$

Then, in Round 052, composing the same 144-vertex controller with the  $\mathbb{F}_3^5$  chart yielded

$$N(12, 5) \geq 34,992.$$

It is therefore natural to view the discovery process as having at least two stages. In the first stage, the route-chart lift construction principle emerged in Round 023 and reached the first complete candidate in Round 036. In the second stage, after the post-completion generalization search encountered a controller bottleneck, the stopping and reorganization in Rounds 047–050 led to the final  $C_4 \times C_6$  controller.

---

## 5. Transcript-based observations on the discovery process

The analysis in this section is process tracing: it describes changes of direction, the role of failures, and the emergence of abstraction as they are observable in the visible transcript. This paper contains no controlled experiment. Accordingly, this section does not claim how much any single element contributed to the discovery.

### 5.1 Record of unsuccessful approaches and changes of direction

The dialogue log contains many unsuccessful search intervals. These intervals include the Round 002 attempt to repair reachability by changing one of the maps assigned to an edge, attempts to modify highly structured candidate graphs (roughly Rounds 004–015), the direction that the LLM tentatively called the “Hashimoto–de Bruijn packet construction” (Rounds 017–018), and the post-completion generalization search (around Rounds 040–046). None of these unsuccessful searches is directly contained in the final construction. We cannot go so far as to say that these failures were exploited and produced the final construction. However, the interpretations the LLM gave of those failures, and the searches it conducted afterward, can be observed from the transcript.

A recurring pattern in the early part of the transcript is caution about improving a nearby candidate. This appears in two forms. The first is after-the-fact local repair: after a defect has been found, one changes a small part of the candidate in order to repair that defect. For instance, in Round 002, the LLM reported that changing one of the maps assigned to an edge in order to make a particular pair reachable could break other reachability relations. The second is a more structured variant: one starts from a highly structured candidate graph and modifies it in a small or uniform way. In several such attempts, the reported candidates were close in a numerical sense, but the remaining obstructions were not localized. The point recorded here is not the specific structure of those candidates, but the reported failure pattern: the obstruction did not appear as a small number of defects that could be removed by a minor adjustment. This observation is recorded only as transcript-level background preceding the later emphasis on explicit covering identities and finite certificate-like conditions.

In the “Hashimoto / de Bruijn type packet” ideas, the obstacle was formulated as the question of how to carry, within a limited number of steps, enough information to distinguish each target. In Round 018, the LLM stated that rather than continuing to repair this direction, one needed first to design how all targets would be distinguished, as in a coded multi-window packet or a covering-array quotient.

In Rounds 040–046, several universal charts were found, but no corresponding controller was found. In Round 047, the LLM stated that one should separate the route-chart principle itself from the ad hoc search for a controller, and withdraw from the latter. In the subsequent Rounds 049–050, the final 144-vertex controller was obtained.

Thus the transcript contains statements in which the LLM verbalized missing conditions in several unsuccessful approaches and, in later searches, appeared to prioritize approaches with a provable covering principle or a finite certificate over the policy of repairing a nearby candidate.

## 5.2 Constraints contained in the initial prompt

The initial prompt in Round 001 contained three kinds of constraints.

1. A clear evaluation goal: obtaining a new record.
2. A novelty constraint demanding ideas not essentially contained in the relevant literature.
3. A constraint suppressing web search during the ideation and construction stages.

In the transcript, this combination appears as pressure toward the search for a new representation rather than a mere re-implementation of known constructions. As noted in §2.3, however, each element of the final construction is clearly connected to existing methods. Relative to existing lift and voltage constructions, the difference emphasized in this paper does not lie in the individual tools themselves. It lies in decomposing those tools into controller-side exact-NB-5 coverage and a fiber-side universal route chart, and then integrating the two parts into a finite certificate for a diameter proof in the degree/diameter problem. This point is discussed in detail in §7.

## 5.3 The user’s meta-level interventions and representative exchanges

In the visible transcript, the author’s general, problem-independent interventions occurred near several junctures at which the level of abstraction of the search changed. The interventions referred to here did not supply components of the final construction. They were general instructions about how to proceed with the search, when to stop, novelty requirements, and requests to explain the prospects of success in advance.

The interventions can be divided broadly into three types. The first is a stop-and-organize type intervention, such as asking the LLM to pause and reconsider the direction. The second is a new-principle-focused intervention, such as asking it to devise a new principle or to concentrate on mathematically new ideas. The third is an a-priori-prospect intervention, such as asking it to adopt a direction for which there is a mathematical reason to expect success. All are general research instructions transferable to other search problems.

The following table summarizes exchanges that are especially important for reading the discovery process. User utterances are quoted briefly, while the GPT side is summarized.

| Round | User prompt (excerpt)                                                                                            | Summary of GPT response                                                                                                                                             | Author’s comment                                                    |
|-------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 001   | “an idea not essentially contained in the literature”; “do not use web search, in principle”                     | After surveying the table of known records, states that it will try a construction that is not an extension of existing methods                                     | Gave general constraints on the search as an initial condition      |
| 003   | “I do not feel that a good graph can be obtained by local improvement”                                           | Acknowledges that random lift repair is weak as the main strategy for a large new record, and switches to a construction whose diameter proof closes without repair | Appears to have curbed the drift toward local repair                |
| 017   | also prohibits “simply trying constructions from methods or concepts a mathematician would think of immediately” | States that it will “rebuild the construction principle itself, rather than improving candidates,” and makes the waste of the distance itself the object of design  | A point at which the novelty constraint was explicitly strengthened |

| Round   | User prompt (excerpt)                                                                                                                                      | Summary of GPT response                                                                                                                                                                       | Author’s comment                                                                                        |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 022     | requests concentration on “ideas with a reason to expect efficient connectivity in advance”                                                                | Discards the “projective rank-controller direction” as lacking an a priori basis, and says that a new principle with an explicit covering identity is needed                                  | A point at which the evaluation axis moved from mere near misses to a provable covering principle       |
| 023     | “please devise a new principle”                                                                                                                            | Proposes the controllable route-chart lift, explaining that a single route together with $q^5$ controls covers the whole fiber                                                                | The point at which the abstraction underlying the final construction was first made explicit            |
| 026     | “For the current 58-vertex controller problem, existing ideas may be used, provided that there is a mathematical reason to expect success.”                | Treats existing voltage-lift ideas as admissible for the isolated controller subproblem, but only under an a-priori-success criterion                                                         | Clarified the boundary between the novelty constraint and the controlled use of existing tools          |
| 036     | “please continue”                                                                                                                                          | From a 132-vertex exact-NB-5 controller and an $\mathbb{F}_4^5$ chart, obtains the first complete candidate (16, 5, 135, 168)                                                                 | A cautionary example showing that large progress can occur even after a simple continuation instruction |
| 047     | “please pause and review the direction”                                                                                                                    | Summarizes that the generalization search is drifting toward trial-and-error controller search, and proposes a “sumset-certified controller”                                                  | A point at which the search after the first complete candidate was re-abstracted                        |
| 049–050 | A renewed request in Round 049 to avoid “methods with no mathematical reason to expect success,” followed by a brief continuation instruction in Round 050 | Focuses the controller search on a $C_4 \times C_6$ -based voltage-lift design, and in Round 050 obtains the final 144-vertex exact-NB-5 controller, yielding the 147,456-vertex construction | The final controller update                                                                             |

For Rounds 049–050, the constraint introduced in Round 049 was to narrow the search to directions with an a priori prospect of success. This constraint was continued by the brief continuation instruction in Round 050. For this reason, the paper treats Rounds 049–050 as a single unit of progress. The analysis treats Round 023, Round 036, and Rounds 049–050, together with their surrounding exchanges, as focal points. These are the rounds in which the route-chart principle, the first complete candidate, and the final controller update respectively appeared.

Among these, Round 023 and Rounds 049–050 are especially salient in relation to the user’s meta-level interventions. In Round 023, an explicit intervention requesting a “new principle” immediately preceded the response in which the route-chart lift first appeared. This route-chart lift became the core of the final construction. In Rounds 049–050, the post-completion generalization search had already encountered several controller bottlenecks. The request in Round 049 to concentrate on “ideas for which there is an a priori reason to expect that they have the property currently desired” was followed by a focus on the  $C_4 \times C_6$  type construction used in the final construction. The brief

continuation instruction in Round 050 then led to the  $C_4 \times C_6$  controller.

By contrast, the first complete candidate in Round 036 was obtained after eight consecutive rounds of brief continuation instructions, including Round 036 itself.

At the time of Rounds 049–050, however, a separate ChatGPT session was attempting in parallel to generalize the result around Round 036. Therefore, although the final idea itself was not obtained in the separate session, this paper does not exclude the possibility that verification results, failure information, or the search state in that separate session indirectly contributed to the progress of Rounds 049–050.

The user’s prompts can be broadly classified into three groups. The first group set the goal of the search, as in Rounds 001, 039, and 052. The second requested an explanation of results or output to a separate file, as in Rounds 028, 037, 038, 042, and 051. The third requested continuation of the search.

There were 44 continuation requests. Of these, 31 (about 70%) added no new mathematical constraint or policy and simply prompted continuation of the current search with phrases such as “please proceed” or “please continue.” These occurred in Rounds 002, 005, 006, 008, 010, 012, 013, 014, 015, 016, 018, 019, 020, 021, 024, 029, 030, 031, 032, 033, 034, 035, 036, 040, 041, 043, 044, 045, 046, 048, and 050. The remaining 13 continuation requests added meta-interventions involving the addition, modification, or relaxation of constraints.

---

## 6. Limitations

This paper reports a single case. It does not show that ChatGPT can in general perform similar mathematical discovery with high probability, nor that the present method is generally effective. Evaluating success rate of this strategy or the effectiveness of meta-level interventions would require multiple independent runs and ablation studies.

The analysis in this paper is limited to the visible transcript and the artifacts. We do not observe the LLM’s internal reasoning, hidden chain-of-thought, or the sources actually referenced by the model. Thus, regarding what the LLM relied on and how it reasoned, we can only infer from its output proposals, code, and changes of direction.

Finally, the scope of the process analyzed in this paper should be distinguished from the preparation of Appendix A. Appendix A was not generated solely by the LLM used in the session analyzed here. It was prepared by the author from files generated in the observed session, together with later LLM-assisted drafts produced using Claude Code Opus 4.7, Claude Code Opus 4.8, and ChatGPT Pro, and was then edited by the author. The process analyzed in this paper is the process leading to the construction of the solution, including the generation of the concrete graph edge lists included in the supplementary materials. It does not include the later process of organizing the mathematical justification of that construction into the proof exposition given in Appendix A.

## 7. Discussion

### 7.1 Novelty of the graph construction method

The existing framework closest to the present construction is that of voltage graphs and graph coverings. In a voltage assignment, one assigns a group element to each directed edge of the base graph, and the product or sum along a walk describes the movement of the fiber coordinate in the lift. In the degree/diameter problem as well, voltage-assignment methods for constructing a large lift from a small base graph have been widely used [BrankovicEtAl1998a, LozSiran2008]. The controller certificate in this paper belongs to the same lineage as this standard voltage-lift idea. Indeed, the 144-vertex controller is the  $C_4 \times C_6$  voltage lift of a 6-vertex multibase, and its exact-NB-5 property is verified by checking that the voltage sums of nonbacktracking walks of length 5 cover all target voltages.

The final route-chart lift, however, is not the ordinary group-valued voltage lift of the original controller  $C$ . In an ordinary voltage lift, a single directed edge of the base determines a single permutation, or a single group displacement, on the fiber. In this paper, by contrast, a controller edge with label  $t$  is associated with the  $q$  affine maps

$$x \mapsto A_t x + \lambda b_t \quad (\lambda \in \mathbb{F}_q).$$

Formally, this can be regarded as a permutation voltage lift over the multigraph obtained by expanding each edge of the controller into  $q$  control-indexed parallel edges [GrossTucker2001, DalfóEtAl2021]. From this point of view, the novelty of the construction lies in choosing the affine permutation voltages so that every reduced route of length 5 parametrizes the entire fiber, and in verifying this fact by a finite list of rank certificates.

Concretely, writing the label sequence of a fixed controller route as  $w = t_1 \cdots t_5$ , the endpoint of the fiber coordinate is given by

$$x_5 = A_w x_0 + M_w(\lambda_1, \dots, \lambda_5)^T.$$

Thus the nonsingularity of  $M_w$  means that the single controller route has a unique control sequence reaching any prescribed target fiber coordinate. This route-wise controllability condition separates the diameter check for the huge final graph into the controller-side exact route coverage and the fiber-side rank condition over all reduced words. This separation is central to the certificate design in this paper.

### 7.2 Decomposition of search execution and search control

The search strategy used by the author was to have the LLM carry out candidate generation and mathematical reasoning, while the author controlled the hierarchy of the search, stopping decisions, and acceptance criteria through general interventions.

This can be understood as a working hypothesis that divides the search into an “execution process,” which advances the search within the problem, and a “control process,” which adjusts the granularity and acceptance criteria of the search from outside the problem. This paper does not test the effectiveness of this decomposition. It records instances, including Round 023 and Rounds 049–050, where control-type interventions and increases in the level of abstraction occur in close proximity.

### 7.3 Relation to recent cases

Apart from the system-oriented methods listed in §1, there have recently been reports of research-level mathematical results obtained solely through dialogue with a browser-accessible LLM ordinarily available to researchers, without a dedicated search engine, formalizer, or orchestration [Bubeck2025, Salim2025, Verbeken2026, Gowers2026, Howlett2026]. In particular, [Verbeken2026] is a single-case study that recorded and analyzed the resolution of an open conjecture by a consumer LLM across multiple dialogue sessions. It also released the related artifacts. Methodologically, it is closest to Contribution 2 of this paper.

Several reports suggest that browser-accessible LLMs can contribute to new mathematical results. What is especially relevant here is the nature of human involvement. These reports are not uniform in the role assigned to the human participant. At least within the released and recorded materials, some describe processes in which mathematically concrete directions or pointers were supplied through prompts before the final result was reached [Salim2025, Verbeken2026]. Others explicitly state that the human made no essential mathematical contribution, and that the human role was limited to guidance such as posing the problem, continuing the search, and requesting confirmation of correctness [Gowers2026]. The materials released in [Gowers2026], however, consist mainly of artifacts and a narrative record, rather than a long verbatim transcript of the kind released with this paper. Moreover, that case is described as a relatively short exchange on the order of minutes, so the time scale of the record differs from the 52-round search treated in this paper.

With respect to the nature of human involvement, the case reported here is closer to the latter type than to the former. The author’s contribution was concentrated on general control not specific to the degree/diameter problem. This control included demanding novelty, stopping and re-abstracting, and requesting explanations of the prospects of success in advance. It did not consist in supplying concrete components of the construction (§4, §5).

The difference from [Gowers2026] lies instead in the form and time scale of the record. This paper releases a visible transcript spanning 52 rounds, including unsuccessful search intervals. It also records, as process tracing, how structured meta-level interventions corresponded in time to changes in the level of abstraction. These interventions include stopping and re-abstracting, requesting novelty, and asking for prospects of success in advance.

Thus, beyond the absence of problem-specific proof or construction hints from the author, the distinctive features of this paper are its time scale and its mode of recording the interaction.

### 7.4 Future work

This work suggests three perspectives to consider when LLM agents are used for automatic search on general problems. The first concerns the role of search control. In the present case, this role was played by the human author. It remains to be studied how well the search would proceed if this role were delegated instead to another LLM agent.

The second concerns the separation between search execution and search control. One possible design is to let a single LLM agent both advance the search within the problem and adjust the granularity and acceptance criteria of the search. Another is to separate these two processes explicitly and assign them to different LLM agents. It remains unclear what theoretical and practical differences would arise between these designs.

The third concerns the comparison between lightweight control loops and dedicated systems. When

a powerful foundation model is guided only by a lightweight control loop, either a human controller or a second LLM, it is natural to ask how close such a workflow can come to the level attainable by dedicated systems such as FunSearch [RomeraParedes2024] or AlphaProof [Hubert2026]. It also remains to understand in which situations the lightweight-control approach and the dedicated-system approach are respectively preferable.

This paper provides only a single data point showing that a mathematical result can be reached through a lightweight workflow without a dedicated search system. It also remains unclear how far the observations generalize to other fields requiring exploratory thinking. These questions are left for future work.

---

## 8. Conclusion

This paper presented a construction proving the new lower bounds

$$N(12, 5) \geq 34,992, \quad N(16, 5) \geq 147,456$$

for the degree/diameter problem. It also gave an account of the approximately six-day search process through which the construction was discovered in dialogue with ChatGPT via its standard web interface.

The transcript contains several points at which the LLM’s responses verbalize lessons drawn from unsuccessful approaches. It also contains several points at which the level of abstraction of the search appears to change after user prompts imposing general constraints. Evaluating the effectiveness of such control and constraints, and the extent to which similar methods can handle difficult problems more broadly, remains a topic for future work.

---

## Appendix A. Mathematical construction and proofs

This appendix collects the mathematical constructions needed to prove Theorem 2.1.

### A.1 Multigraphs and nonbacktracking walks

For a finite set  $V$ , write

$$\binom{V}{2} := \{\{u, v\} : u, v \in V, u \neq v\}.$$

This is the set of all two-element subsets of  $V$ .

**Definition A.1.** A *loopless multigraph* is a triple  $B = (V, E, \partial)$  where  $V$  and  $E$  are finite sets and  $\partial : E \rightarrow \binom{V}{2}$  is a map. When  $\partial(e) = \{u, v\}$  for  $e \in E$ , we say that  $e$  is an edge joining  $u$  and  $v$ . Distinct edges  $e, e' \in E$  with the same endpoint set  $\{u, v\}$  are allowed. On the other hand, by the definition of  $\binom{V}{2}$ ,  $\partial(e)$  always consists of two distinct vertices, so there are no loops.

**Definition A.2.** A walk of length  $\ell$  in  $B = (V, E, \partial)$  is a sequence

$$P = (v_0, e_1, v_1, e_2, \dots, e_\ell, v_\ell)$$

with  $v_i \in V$ ,  $e_i \in E$  and  $\partial(e_i) = \{v_{i-1}, v_i\}$  for all  $1 \leq i \leq \ell$ . This walk is *nonbacktracking* (NB) if

$$e_{i+1} \neq e_i \quad (1 \leq i < \ell)$$

is satisfied. For simple graphs, this coincides with the usual condition  $v_{i+1} \neq v_{i-1}$ .

## A.2 Voltage lifts and the controller

Voltage graphs and regular coverings are standard construction tools. In this paper we use voltage assignments with values in a finite abelian group [GrossTucker2001]. In the degree/diameter problem too, the method of giving a voltage assignment to a small base graph to build a large lift, and reducing the diameter condition to a condition on walks in the base graph and net voltages, has been used [BrankovicEtAl1998a, LozSiran2008].

Let  $B = (V, E, \partial)$  be a loopless multigraph. For each edge  $e \in E$ , with  $\partial(e) = \{u, v\}$ , consider the two orientations  $(u, e, v)$  and  $(v, e, u)$ . We call these the directed edges of  $B$ , and write  $\vec{E}(B)$  for the set of all directed edges. We write the reverse of a directed edge  $\vec{e} = (u, e, v)$  as  $\bar{\vec{e}} = (v, e, u)$ .

**Definition A.3.** Let  $H$  be a finite abelian group. An  $H$ -valued voltage assignment is a map

$$\gamma : \vec{E}(B) \rightarrow H$$

satisfying, for every directed edge  $\vec{e}$ ,

$$\gamma(\bar{\vec{e}}) = -\gamma(\vec{e}).$$

The group element  $\gamma(\vec{e})$  assigned to a directed edge  $\vec{e}$  is called the voltage of that directed edge.

The regular lift  $B^\gamma$  of a voltage-equipped multigraph  $(B, \gamma)$  is the following, not necessarily simple, undirected multigraph. The vertex set is

$$V(B^\gamma) = V(B) \times H.$$

For each edge  $e \in E(B)$ , choose one of its two orientations, say  $\vec{e} = (u, e, v)$ . For each  $h \in H$ , add an undirected edge, with edge id  $(e, h)$ , joining

$$(u, h) \quad \text{and} \quad (v, h + \gamma(\vec{e})).$$

This definition is independent of the chosen orientation. Indeed, if the reverse orientation is used and  $h' = h + \gamma(\bar{\vec{e}})$ , then

$$h = h' + \gamma(\vec{e}),$$

so the same undirected edge is obtained.

Given a walk  $P = (v_0, e_1, v_1, \dots, e_\ell, v_\ell)$  and an initial value  $h_0 \in H$ , let the directed edge of the  $i$ -th step be  $\vec{e}_i = (v_{i-1}, e_i, v_i)$  and set

$$h_i = h_{i-1} + \gamma(\vec{e}_i).$$

Then  $h_\ell - h_0 = \sum_{i=1}^{\ell} \gamma(\vec{e}_i)$  is called the voltage sum of  $P$ .

**Lemma A.4.** Let  $(B, \gamma)$  be an  $H$ -valued voltage-equipped loopless multigraph, and set  $C = B^\gamma$ . Assume that for some  $s \geq 1$  the following holds.

$$\forall i, j \in V(B), \quad \forall h \in H : \text{ there is a length-}s \text{ NB walk } i \rightarrow j \text{ in } B \text{ with voltage sum } h. \quad (\text{A.0})$$

Then, for any  $(c, d) \in V(C)^2$ , there exists a nonbacktracking walk of length exactly  $s$  from  $c$  to  $d$  in  $C$ .

**Proof.** Write  $c = (i, h_1)$ ,  $d = (j, h_2)$ . Applying (A.0) to  $h = h_2 - h_1$ , there is a length- $s$  NB walk  $P$  in  $B$  from  $i$  to  $j$  whose voltage sum is  $h_2 - h_1$ . Lifting  $P$  from the initial value  $h_1$  by the rule above, the second component of the endpoint is  $h_1 + (h_2 - h_1) = h_2$ . The NB condition is the condition of not immediately returning along the same edge id, and is therefore preserved after lifting. Hence a length- $s$  NB walk from  $c$  to  $d$  is obtained in  $C$ .  $\square$

A simple graph  $C$  with the above property is called an **exact-NB- $s$  controller**.

### A.3 Symmetric alphabets and inverse-consistent labelings

**Definition A.5.** When a map  $\iota : S \rightarrow S$  on a finite set  $S$  satisfies  $\iota(\iota(t)) = t$  for all  $t \in S$ ,  $\iota$  is called an involution. If furthermore  $\iota(t) \neq t$  holds for all  $t \in S$ , then  $\iota$  is said to be fixed-point-free. In this paper, a finite set  $S$  equipped with a fixed-point-free involution is called a symmetric alphabet, and we write  $\iota(t)$  as  $t^{-1}$ .

A word  $w = t_1 \cdots t_s$  of length  $s$  is *reduced* if

$$t_{i+1} \neq t_i^{-1} \quad (1 \leq i < s).$$

Let  $C$  be a  $\rho$ -regular simple undirected graph and  $S$  a symmetric alphabet with  $|S| = \rho$ . Write  $\vec{E}(C)$  for the set of all directed edges of  $C$ . A map  $\ell : \vec{E}(C) \rightarrow S$  is an *inverse-consistent labeling* if it satisfies the following two conditions.

1. For each vertex  $u \in V(C)$ , the labels of the  $\rho$  directed edges leaving  $u$  take each element of  $S$  exactly once.
2. For any directed edge  $u \rightarrow v$ ,  $\ell(v \rightarrow u) = \ell(u \rightarrow v)^{-1}$ .

Then the label sequence of any NB walk in  $C$  is a reduced word. Let two consecutive directed edges of a walk be  $c_{i-1} \rightarrow c_i$  and  $c_i \rightarrow c_{i+1}$ , and let the label of the former be  $t$ . By condition 2, the label of the reverse directed edge  $c_i \rightarrow c_{i-1}$  is  $t^{-1}$ . By condition 1, there is exactly one directed edge leaving  $c_i$  with label  $t^{-1}$ , so if the next label is  $t^{-1}$  then  $c_i \rightarrow c_{i+1}$  is the directed edge that reverses the previous edge. This contradicts the NB condition.

By Petersen's 2-factor theorem [Petersen1891], an even-regular simple undirected graph admits a 2-factor decomposition. Assign a distinct inverse pair  $\{t, t^{-1}\}$  of  $S$  to each 2-factor. Choose an orientation of each cycle, and assign  $t$  to the forward directed edges and  $t^{-1}$  to the backward directed edges. Then, at each vertex, the pair  $\{t, t^{-1}\}$  appears once from each 2-factor, and an inverse-consistent labeling is obtained. In this paper we take two inverse pairs, namely  $S = \{a, a^{-1}, b, b^{-1}\}$ .

### A.4 Route charts and chart lifts

Let  $S$  be a symmetric alphabet,  $\mathbb{F}_q$  a finite field, and  $s \geq 1$ .

**Definition A.6.** A *route chart* on  $\mathbb{F}_q^s$  is an assignment, to each  $t \in S$ , of

$$A_t \in \text{GL}_s(\mathbb{F}_q), \quad b_t \in \mathbb{F}_q^s,$$

satisfying, for all  $t \in S$ ,

$$A_{t^{-1}} = A_t^{-1}, \quad b_{t^{-1}} = A_t^{-1} b_t \tag{A.1}$$

For a word  $w = t_1 \cdots t_s$  of length  $s$ , set

$$A_w := A_{t_s} A_{t_{s-1}} \cdots A_{t_1}.$$

For a reduced word  $w = t_1 \cdots t_s$  of length  $s$ , define the controllability matrix by

$$M_w = [A_{t_s} \cdots A_{t_2} b_{t_1} \mid A_{t_s} \cdots A_{t_3} b_{t_2} \mid \cdots \mid A_{t_s} b_{t_{s-1}} \mid b_{t_s}] \in M_s(\mathbb{F}_q) \quad (\text{A.2})$$

A route chart is *universal* if  $M_w$  is nonsingular for every reduced word  $w$  of length  $s$ .

**Definition A.7.** Let  $C$  be a  $\rho$ -regular simple undirected graph,  $S$  a symmetric alphabet with  $|S| = \rho$ ,  $\ell : \vec{E}(C) \rightarrow S$  an inverse-consistent labeling, and  $(A_t, b_t)_{t \in S}$  a route chart on  $\mathbb{F}_q^s$ . Define the route-chart lift graph

$$G = G(C, S, \ell, A, b)$$

as follows. The vertex set is

$$V(G) = V(C) \times \mathbb{F}_q^s.$$

For an undirected edge  $\{u, v\}$  of  $C$  with  $\ell(u \rightarrow v) = t$ , for each  $x \in \mathbb{F}_q^s$  and each  $\lambda \in \mathbb{F}_q$ , add the undirected edge joining the two vertices

$$(u, x), \quad (v, A_t x + \lambda b_t).$$

We show that this rule is independent of the orientation chosen for the undirected edge  $\{u, v\}$ . Let the label of the direction  $u \rightarrow v$  be  $t$ . For arbitrary  $x \in \mathbb{F}_q^s$  and  $\lambda \in \mathbb{F}_q$ , set

$$y = A_t x + \lambda b_t.$$

Then the rule using the orientation  $u \rightarrow v$  adds the edge joining  $(u, x)$  and  $(v, y)$ .

The reverse direction  $v \rightarrow u$  has label  $t^{-1}$ . By (A.1), for any  $\mu \in \mathbb{F}_q$ ,

$$A_{t^{-1}} y + \mu b_{t^{-1}} = A_t^{-1} (A_t x + \lambda b_t) + \mu A_t^{-1} b_t = x + (\lambda + \mu) A_t^{-1} b_t.$$

Taking  $\mu = -\lambda$  gives  $A_{t^{-1}} y + \mu b_{t^{-1}} = x$ . Thus the reverse orientation adds the same edge  $\{(v, y), (u, x)\}$ . Since  $\lambda \mapsto -\lambda$  is a bijection of  $\mathbb{F}_q$ , the two orientations give the same family of edges, and the edge-adding rule is well-defined for the undirected edge. In particular, for any directed edge  $u \rightarrow v$  of  $C$  (label  $t$ ),

$$(u, x) \text{ and } (v, y) \text{ are adjacent} \iff \exists \lambda \in \mathbb{F}_q : y = A_t x + \lambda b_t.$$

**Lemma A.8.** With the above notation, suppose the route chart is universal. Then the following hold.

1.  $G$  is a simple undirected graph.
2.  $G$  is  $\rho q$ -regular.
3. If  $C$  is an exact-NB- $s$  controller then  $\text{diam}(G) \leq s$ .
4.  $|V(G)| = |V(C)|q^s$ .

**Proof.** The graph is undirected by Definition A.7. We first prove simplicity and the degree. Since  $C$  is simple, for a directed edge  $u \rightarrow v$  of  $C$  we always have  $u \neq v$ . Hence  $G$  has no loops. By universality, each  $b_t$  is nonzero. Indeed, the word  $t^s$  is reduced, and the last column of the corresponding  $M_{t^s}$  is  $b_t$ ; if  $M_{t^s}$  is nonsingular then  $b_t \neq 0$ .

Consider a fixed vertex  $(u, x) \in V(G)$ . There are  $\rho$  directed edges leaving  $u$ , and for each, choosing  $\lambda \in \mathbb{F}_q$  yields the candidate neighbor

$$(v, A_t x + \lambda b_t).$$

By the argument at the end of Definition A.7, these are all the neighbors of  $(u, x) \in V(G)$ . Distinct directed edges go to distinct controller vertices  $v$  because  $C$  is simple. For the same directed edge, if

$$A_t x + \lambda b_t = A_t x + \lambda' b_t,$$

then  $(\lambda - \lambda')b_t = 0$ , and from  $b_t \neq 0$  we get  $\lambda = \lambda'$ . Hence  $(u, x)$  has exactly  $\rho q$  distinct neighbors. Therefore  $G$  is simple and  $\rho q$ -regular.

Next we show the diameter. Take any  $(c, x), (d, y) \in V(G)$ . Since  $C$  is exact-NB- $s$ , there is an NB walk of length  $s$  in  $C$ ,

$$c = c_0, c_1, \dots, c_s = d.$$

Let the label of  $c_{i-1} \rightarrow c_i$  be  $t_i$ . As shown in the previous subsection, under the inverse-consistent labeling, the label sequence  $w = t_1 \dots t_s$  of an NB walk is a reduced word.

Choose control parameters  $\lambda_1, \dots, \lambda_s \in \mathbb{F}_q$  along this walk. Starting the fiber coordinate at  $x_0 = x$  and setting

$$x_i = A_{t_i} x_{i-1} + \lambda_i b_{t_i} \quad (1 \leq i \leq s),$$

we obtain inductively

$$x_s = A_w x + M_w \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_s \end{pmatrix}.$$

By universality  $M_w$  is nonsingular, so the linear equation

$$M_w \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_s \end{pmatrix} = y - A_w x$$

has a solution. Using this solution, we obtain a walk of length  $s$  from  $(c, x)$  to  $(d, y)$ . Hence  $\text{diam}(G) \leq s$ . The vertex count follows immediately from the definition.  $\square$

**Remark.** The route-chart lift of Definition A.7 can also be described as a permutation voltage lift over the expanded base multigraph. That is, replace each directed edge of  $C$  with label  $t$  by  $q$  directed edges indexed by  $\lambda \in \mathbb{F}_q$ , and assign to that directed edge the permutation of  $\mathbb{F}_q^s$

$$\pi_{t,\lambda}(x) = A_t x + \lambda b_t.$$

By condition (A.1), replacing the control parameter  $\lambda$  by  $-\lambda$  makes the affine map on the reverse directed edge the inverse permutation. Hence the route-chart lift is an instance of a graph lift by a permutation voltage assignment [GrossTucker2001, DalfoEtAl2021]. In this paper, in addition to this permutation lift, we impose the nonsingularity of the controllability matrix for all reduced routes of length  $s$ , thereby reducing the diameter proof to a finite list of rank certificates.

## A.5 The 144-vertex controller

Let the vertex set of  $B$  be  $\{0, 1, 2, 3, 4, 5\}$  and the voltage group be

$$H = C_4 \times C_6 = \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}.$$

Each edge is as in Table A.1, with the orientation  $u \rightarrow v$  in the table taken as the forward direction.

**Table A.1.** Base multigraph  $B$  and voltage assignment.

| edge id | $u$ | $v$ | $C_4$ component | $C_6$ component |
|---------|-----|-----|-----------------|-----------------|
| 0       | 0   | 2   | 1               | 0               |
| 1       | 0   | 3   | 2               | 4               |
| 2       | 0   | 4   | 2               | 0               |
| 3       | 0   | 4   | 2               | 5               |
| 4       | 1   | 2   | 1               | 2               |
| 5       | 1   | 3   | 3               | 5               |
| 6       | 1   | 5   | 1               | 1               |
| 7       | 1   | 5   | 1               | 0               |
| 8       | 2   | 4   | 2               | 2               |
| 9       | 2   | 5   | 1               | 5               |
| 10      | 3   | 4   | 1               | 5               |
| 11      | 3   | 5   | 3               | 1               |

Each vertex occurs four times in Table A.1, and there are no loops. Hence  $B$  is a loopless 4-regular multigraph. Setting  $C := B^\gamma$ ,

$$V(C) = \{0, 1, 2, 3, 4, 5\} \times C_4 \times C_6,$$

so  $|V(C)| = 6 \cdot 4 \cdot 6 = 144$ .

**Proposition A.9.**  $C$  is a simple 4-regular connected graph and an exact-NB-5 controller.

**Proof.** Regularity follows from the definition of the regular lift and the 4-regularity of  $B$ . As for simplicity, since  $B$  has no loop, the lift has no loop. The only parallel pairs of base edges are edges 2 and 3, and edges 6 and 7; in each pair, the voltages are different. For two base edges joining the same two base vertices to give the same lift edge, their voltages must be equal, so no multiple edge arises.

The base multigraph  $B$  is connected by inspection of Table A.1. Connectivity of the lift follows from the fact that the closed-walk voltages of the base generate all of  $H = C_4 \times C_6$ . For example, the voltage of

$$0 \xrightarrow{e_3} 4 \xrightarrow{e_2^{-1}} 0$$

is  $(2, 5) - (2, 0) = (0, 5)$ , which generates the  $C_6$  component. Also, the voltage of

$$5 \xrightarrow{e_6^{-1}} 1 \xrightarrow{e_4} 2 \xrightarrow{e_9} 5$$

is  $-(1, 1) + (1, 2) + (1, 5) = (1, 0)$ , which generates the  $C_4$  component. Hence  $C$  is connected.

For the exact-NB-5 property, by Lemma A.4 it suffices to verify that, for any  $i, j \in \{0, \dots, 5\}$  and any  $h \in C_4 \times C_6$ , there is a length-5 NB walk  $i \rightarrow j$  in  $B$  with voltage sum  $h$ . The number of length-5 NB walks from each starting vertex of  $B$  is at most  $4 \cdot 3^4 = 324$ , and over all starting vertices it is at most  $6 \cdot 324 = 1944$ , so a complete enumeration is possible. The supplementary `scripts/verify_candidate.py` enumerates these by DFS and verifies that all 36 ordered base pairs and all 24 voltage classes are covered.  $\square$

## A.6 Universal route charts for $s = 5$

In this section we set  $S = \{0, 1, 2, 3\}$  with the correspondence  $0 = a$ ,  $1 = a^{-1}$ ,  $2 = b$ ,  $3 = b^{-1}$ . The involution is  $0 \leftrightarrow 1$ ,  $2 \leftrightarrow 3$ . There are  $4 \cdot 3^4 = 324$  reduced words of length 5.

### A.6.1 Chart on $\mathbb{F}_3^5$

Let  $\mathbb{F}_3 = \{0, 1, 2\}$ . We use the following matrices and vectors.

$$A_0 = \begin{pmatrix} 1 & 2 & 0 & 2 & 0 \\ 1 & 2 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 0 \\ 2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 2 & 0 & 1 \end{pmatrix}, \quad A_1 = \begin{pmatrix} 0 & 0 & 0 & 2 & 0 \\ 2 & 2 & 1 & 2 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{pmatrix},$$

$$A_2 = \begin{pmatrix} 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 2 & 2 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 0 \\ 2 & 2 & 1 & 1 & 1 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 2 & 0 \\ 0 & 1 & 1 & 2 & 0 \\ 2 & 1 & 2 & 1 & 1 \end{pmatrix},$$

$$b_0 = \begin{pmatrix} 2 \\ 2 \\ 2 \\ 0 \\ 0 \end{pmatrix}, \quad b_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad b_2 = \begin{pmatrix} 2 \\ 1 \\ 2 \\ 0 \\ 2 \end{pmatrix}, \quad b_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 2 \end{pmatrix}.$$

**Proposition A.10.** The correspondence  $(A_t, b_t)_{t \in S}$  is a universal route chart on  $\mathbb{F}_3^5$ .

**Proof.** The supplementary `scripts/12_5_34992/verify_candidate.py` verifies the following with arithmetic over  $\mathbb{F}_3$ .

1.  $A_0 A_1 = A_1 A_0 = I_5$ ,  $A_2 A_3 = A_3 A_2 = I_5$ .
2.  $A_1 b_0 = b_1$ ,  $A_0 b_1 = b_0$ ,  $A_3 b_2 = b_3$ ,  $A_2 b_3 = b_2$ .
3. For all 324 reduced words  $w$  of length 5, the matrix  $M_w$  of (A.2) has rank 5.

These are exactly the conditions of Definition A.6 and universality.  $\square$

### A.6.2 Chart on $\mathbb{F}_4^5$

Let  $\mathbb{F}_4 = \mathbb{F}_2[\alpha]/(\alpha^2 + \alpha + 1)$ , with the encoding  $0 = 0$ ,  $1 = 1$ ,  $2 = \alpha$ ,  $3 = \alpha + 1$ . We use the following matrices and vectors.

$$\begin{aligned}
A_0 &= \begin{pmatrix} 2 & 0 & 1 & 2 & 0 \\ 1 & 2 & 2 & 0 & 0 \\ 3 & 0 & 0 & 3 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 2 & 1 & 2 & 2 & 2 \end{pmatrix}, & A_1 &= \begin{pmatrix} 2 & 0 & 3 & 2 & 0 \\ 0 & 3 & 1 & 1 & 0 \\ 1 & 0 & 3 & 0 & 0 \\ 2 & 0 & 1 & 2 & 0 \\ 1 & 2 & 2 & 3 & 3 \end{pmatrix}, \\
A_2 &= \begin{pmatrix} 3 & 1 & 3 & 2 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ 3 & 2 & 2 & 0 & 2 \end{pmatrix}, & A_3 &= \begin{pmatrix} 0 & 3 & 0 & 3 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 2 & 3 & 2 & 0 \\ 3 & 1 & 1 & 2 & 0 \\ 0 & 2 & 3 & 3 & 3 \end{pmatrix}, \\
b_0 &= \begin{pmatrix} 2 \\ 1 \\ 2 \\ 0 \\ 2 \end{pmatrix}, & b_1 &= \begin{pmatrix} 2 \\ 1 \\ 3 \\ 1 \\ 2 \end{pmatrix}, & b_2 &= \begin{pmatrix} 2 \\ 1 \\ 0 \\ 2 \\ 2 \end{pmatrix}, & b_3 &= \begin{pmatrix} 2 \\ 1 \\ 1 \\ 3 \\ 2 \end{pmatrix}.
\end{aligned}$$

**Proposition A.11.** The correspondence  $(A_t, b_t)_{t \in S}$  is a universal route chart on  $\mathbb{F}_4^5$ .

**Proof.** The supplementary `scripts/16_5_147456/verify_candidate.py` verifies, with  $\mathbb{F}_4$  arithmetic using the above encoding, the same three kinds of conditions as in Proposition A.10. That is, the inverse relations (A.1) and the rank-5 condition for all 324 controllability matrices hold.  $\square$

## A.7 Proof of the main theorem

**Theorem A.12.** The following inequalities hold:

$$N(12, 5) \geq 34,992, \quad N(16, 5) \geq 147,456.$$

**Proof.** Fix the 144-vertex controller  $C$  of §A.5 and the inverse-consistent labeling  $\ell$  of §A.3.

First let  $q = 3$  and use the universal route chart of §A.6.1. By Proposition A.9,  $C$  is exact-NB-5, and by Proposition A.10 the chart is universal. By Lemma A.8,

$$G_3 = G(C, S, \ell, A, b)$$

is simple  $4q = 12$ -regular with  $\text{diam}(G_3) \leq 5$ , and satisfies

$$|V(G_3)| = |V(C)|q^5 = 144 \cdot 3^5 = 34,992.$$

Similarly, letting  $q = 4$  and using the universal route chart of §A.6.2, by Lemma A.8,

$$G_4 = G(C, S, \ell, A, b)$$

is simple 16-regular with  $\text{diam}(G_4) \leq 5$ , and satisfies

$$|V(G_4)| = 144 \cdot 4^5 = 147,456.$$

Hence the two claimed lower bounds follow.

Furthermore, 34,992 exceeds the (12, 4) Moore upper bound 17,569, and 147,456 exceeds the (16, 4) Moore upper bound 57,857. Hence the diameters of these graphs cannot be at most 4, and in both cases the diameter is exactly 5.  $\square$

## A.8 Verification and reproducibility

Every computer-dependent part of the proof in this paper consists of verifying finitely many conditions. The supplementary materials preserve the construction package produced by the LLM during the dialogue as `route_chart_verified_graphs_package/`. This directory is a frozen provenance artifact, not a cleaned software release. It preserves the process by which the construction was obtained.

Some generating scripts therefore contain unstandardized elements. These include default output paths from the execution environment used during the dialogue and minor differences in the format of the edge-list header. These elements are not part of the mathematical claims of this paper.

The objects used in the mathematical proof are the construction data made explicit in Appendix A and the corresponding finite list of certificate conditions. The natural-language description files inside `route_chart_verified_graphs_package/` are kept as provenance, but the proof is based on the definitions and lemmas of Appendix A and on the verification procedures below.

This package has two parameter directories, `12_5_34992` and `16_5_147456`, and mainly contains the following.

- `construction_details.md`: the LLM-generated explanation of the route-chart lift construction.
- `certificates/*/controller_c4xc6_multibase.txt`: the base data of Table A.1.
- `certificates/12_5_34992/gf3_s5_universal.txt` and `certificates/16_5_147456/gf4_s5_universal.txt`: the chart data of §A.6.
- `scripts/*/verify_candidate.py`: a script that verifies the finite conditions of base voltage coverage, simplicity/regularity/connectivity of the controller, the labeling, and chart universality.
- `scripts/*/generate_final_graph.py`: a script that generates the edge list.
- `graphs/*/final_graph_edges.tsv.gz`: the generated edge list.
- `SHA256SUMS.txt`: SHA-256 checksums of the main files inside `route_chart_verified_graphs_package/`.

`certificates/*/original_README.txt` and `certificates/12_5_34992/original_phase35_report.md` are kept as provenance for explanations produced by the LLM during the process leading to the construction. These descriptions are not a substitute for the proofs in Appendix A.

The verification is divided into three layers.

1. **Certificate-level verification.** Each `scripts/*/verify_candidate.py` verifies the voltage coverage of length-5 nonbacktracking walks on the base multigraph, the simplicity / 4-regularity / connectivity of the regular lift controller, the inverse-consistent labeling, and the rank-5 condition of the controllability matrix for all 324 reduced words of length 5. The expected output has the following form.

```
Base voltage coverage over C4 x C6: OK
Controller: OK; 144 vertices, simple 4-regular, exact labeled NB length-5 coverage
GF(q)^5 universal chart: OK; 324 reduced words checked, min rank 5; inverse-symbol
↔ data verified
Certificate complete: degree 4q, diameter ≤5, vertices = 144*q^5
```

2. **Generation-level identity check.** `scripts/*/generate_final_graph.py` generates the final edge list. To preserve provenance, these generating scripts are, in principle, kept in

the form obtained during the dialogue. Thus, at run time, one should make a working copy and explicitly specify the output path for the generated edge list before comparing it with the bundled file. The top-level `verify_graphs_identical.py` in the supplementary materials verifies that the graph obtained from the generating script and the graph bundled in `route_chart_verified_graphs_package/graphs/*/final_graph_edges.tsv.gz` are identical. The check has two parts. First, it compares the SHA-256 hash of the edge rows, excluding the header. Second, it checks agreement of the edge sets after normalizing the orientation of each edge. The first part verifies agreement of the edge rows in the same order. The second verifies graph-theoretic identity independently of file format or row order.

3. **Graph-level independent check.** `verify_with_networkx.py` is an optional independent verification script that directly reads the bundled edge list and checks the absence of self-loops, the absence of multiple edges, the number of vertices, the number of edges, the regular degree, the handshake equality, and the exact diameter computed by NetworkX. This script does not depend on the natural-language descriptions or the generating scripts inside the LLM-generated package, and uses the edge list itself as its target. Note that the exact diameter computation is an additional check that may take time for large graphs, and the diameter-5 proof of this paper does not depend on this script.

---

## Appendix B. Structure of the supplementary materials

The supplementary materials contain four types of files. These are the visible transcript of the discovery process, the script that converts the conversation log into the visible transcript, the construction package produced by the LLM and preserved as provenance, and the scripts for collation and independent verification.

- `dump_ddp_visible.py`: a script that extracts, from the conversation log obtained via the official ChatGPT download function, only the messages that appear in the browser UI as ordinary user prompt bubbles or assistant answer bubbles. Chain-of-thought, tool calls, tool outputs, reasoning recaps, and system messages are excluded.
- `extracted/DDP_visible_transcript.md`: the visible transcript extracted by `dump_ddp_visible.py`. It targets Rounds 001–052 relevant to the present discovery and verification.
- `route_chart_verified_graphs_package/`: the construction package produced by the LLM during the dialogue, included, in principle, as raw files for provenance preservation. Details are given in A.8 of Appendix A.
- `verify_graphs_identical.py`: a script that verifies that the graph generated from `generate_final_graph.py` and the graph bundled under `route_chart_verified_graphs_package/graphs/` are identical. It checks the SHA-256 of the edge rows excluding the header, and the agreement of the normalized edge set.
- `verify_with_networkx.py`: an independent verification script. It checks the bundled graph for the absence of self-loops, the absence of multiple edges, the number of vertices, the number of edges, the regular degree, and the handshake equality, and also computes the exact diameter with NetworkX. The diameter computation of this script is an optional additional check, and the diameter-5 proof of this paper does not depend on this script.

## References

- [AlonEtAl2026] N. Alon, T. F. Bloom, W. T. Gowers, D. Litt, W. Sawin, A. Shankar, J. Tsimerman, V. Wang, and M. Matchett Wood. Remarks on the disproof of the unit distance conjecture. arXiv:2605.20695, 2026. DOI: 10.48550/arXiv.2605.20695.
- [Bannai1973] E. Bannai and T. Ito. On finite Moore graphs. *Journal of the Faculty of Science, University of Tokyo, Section IA, Mathematics* 20 (1973), 191–208.
- [BermondDelormeFarhi1984] J.-C. Bermond, C. Delorme, and G. Farhi. Large graphs with given degree and diameter II. *Journal of Combinatorial Theory, Series B* 36 (1984), 32–48. DOI: 10.1016/0095-8956(84)90012-1.
- [BermondDelormeQuisquater1986] J.-C. Bermond, C. Delorme, and J.-J. Quisquater. Strategies for interconnection networks: some methods from graph theory. *Journal of Parallel and Distributed Computing* 3 (1986), 433–449. DOI: 10.1016/0743-7315(86)90008-0.
- [BhanNobiliLanger2026] J. Bhan, N. Nobili, and P. Langer. New bounds for Zarankiewicz numbers via reinforced LLM evolutionary search. arXiv:2605.01120, 2026. DOI: 10.48550/arXiv.2605.01120.
- [BrankovicEtAl1998a] L. Branković, M. Miller, J. Plesník, J. Ryan, and J. Širáň. Large graphs with small degree and diameter: A voltage assignment approach. *Australasian Journal of Combinatorics* 18 (1998), 65–76.
- [Bubeck2025] S. Bubeck, C. Coester, R. Eldan, T. Gowers, Y. T. Lee, A. Lupsasca, M. Sawhney, R. Scherrer, M. Sellke, B. K. Spears, D. Unutmaz, K. Weil, S. Yin, and N. Zhivotovskiy. Early science acceleration experiments with GPT-5. arXiv:2511.16072, 2025. DOI: 10.48550/arXiv.2511.16072.
- [Comellas2026] F. Comellas. Table of large graphs with given degree and diameter. arXiv:2406.18994v2, 2026. DOI: 10.48550/arXiv.2406.18994v2.
- [ComellasData2026] F. Comellas. Table of Large Degree/Diameter Graphs. Mendeley Data, Version 11, published 28 January 2026. DOI: 10.17632/d75dzbjd4k.11.
- [DalfóEtAl2021] C. Dalfó, M. A. Fiol, S. Pavlíková, and J. Širáň. Spectra and eigenspaces of arbitrary lifts of graphs. *Journal of Algebraic Combinatorics* 54 (2021), 651–672. DOI: 10.1007/s10801-021-01049-3.
- [DelormeBipartite1985] C. Delorme. Large bipartite graphs with given degree and diameter. *Journal of Graph Theory* 9(3) (1985), 325–334. DOI: 10.1002/jgt.3190090304.
- [Damerell1973] R. M. Damerell. On Moore graphs. *Mathematical Proceedings of the Cambridge Philosophical Society* 74(2) (1973), 227–236. DOI: 10.1017/S0305004100048015.
- [Gowers2026] T. Gowers. A recent experience with ChatGPT 5.5 Pro. Gowers’s Weblog, 8 May 2026. <https://gowers.wordpress.com/2026/05/08/a-recent-experience-with-chatgpt-5-5-pro/>
- [GrossTucker2001] J. L. Gross and T. W. Tucker. *Topological Graph Theory*. Dover, 2001.
- [HoffmanSingleton1960] A. J. Hoffman and R. R. Singleton. On Moore graphs with diameters two and three. *IBM Journal of Research and Development* 4(5) (1960), 497–504. DOI: 10.1147/rd.45.0497.
- [Howlett2026] J. Howlett. An amateur just solved a 60-year-old math problem—by asking AI. *Scientific American*, 24 April 2026.

- [Hubert2026] T. Hubert, R. Mehta, L. Sartran, M. Z. Horváth, G. Žužić, E. Wieser, A. Huang, J. Schrittwieser, Y. Schroecker, H. Masoom, et al. Olympiad-level formal mathematical reasoning with reinforcement learning. *Nature* 651 (2026), 607–613. DOI: 10.1038/s41586-025-09833-y.
- [JuEtAl2026] H. Ju, G. Gao, J. Jiang, B. Wu, Z. Sun, S. Liu, L. Chen, Y. Wang, Y. Wang, Z. Wang, W. He, P. Wu, L. Xiao, R. Liu, B. Dai, and B. Dong. Automated conjecture resolution with formal verification. arXiv:2604.03789, 2026. DOI: 10.48550/arXiv.2604.03789.
- [LozSiran2008] E. Loz and J. Širáň. New record graphs in the degree-diameter problem. *Australasian Journal of Combinatorics* 41 (2008), 63–80.
- [MillerSiran2013] M. Miller and J. Širáň. Moore graphs and beyond: A survey of the degree/diameter problem. *Electronic Journal of Combinatorics*, Dynamic Survey DS14, 2nd edition, 2013. DOI: 10.37236/35.
- [NagdaRaghavanThakurta2026] A. Nagda, P. Raghavan, and A. Thakurta. Reinforced generation of combinatorial structures: Ramsey numbers. arXiv:2603.09172, 2026. DOI: 10.48550/arXiv.2603.09172.
- [OpenAI2026] OpenAI. An OpenAI model has disproved a central conjecture in discrete geometry. 2026. <https://openai.com/index/model-disproves-discrete-geometry-conjecture/>
- [Petersen1891] J. Petersen. Die Theorie der regulären Graphs. *Acta Mathematica* 15 (1891), 193–220.
- [PrzybockiMackeyHeuleSubercaseaux2026] B. Przybocki, J. Mackey, M. J. H. Heule, and B. Subercaseaux. Doubly saturated Ramsey graphs: A case study in computer-assisted mathematical discovery. arXiv:2604.21187, 2026. DOI: 10.48550/arXiv.2604.21187.
- [RomeraParedes2024] B. Romera-Paredes, M. Barekatin, A. Novikov, M. Balog, M. P. Kumar, E. Dupont, F. J. R. Ruiz, J. S. Ellenberg, P. Wang, O. Fawzi, P. Kohli, and A. Fawzi. Mathematical discoveries from program search with large language models. *Nature* 625 (2024), 468–475. DOI: 10.1038/s41586-023-06924-6.
- [Salim2025] A. Salim. Accelerating mathematical research with language models: A case study of an interaction with GPT-5-Pro on a convex analysis problem. arXiv:2510.26647, 2025. DOI: 10.48550/arXiv.2510.26647.
- [Trinh2024] T. H. Trinh, Y. Wu, Q. V. Le, H. He, and T. Luong. Solving olympiad geometry without human demonstrations. *Nature* 625 (2024), 476–482. DOI: 10.1038/s41586-023-06747-5.
- [TsoukalasEtAl2026] G. Tsoukalas, A. Kovsharov, S. Shirobokov, A. Surina, M. Firsching, G. Bérczi, F. J. R. Ruiz, A. Suggala, A. Z. Wagner, E. Wieser, L. Yu, A. Huang, M. Z. Horváth, A. Ferraiuolo, H. Michalewski, C. Grosu, T. Hubert, M. Balog, P. Kohli, and S. Chaudhuri. Advancing mathematics research with AI-driven formal proof search. arXiv:2605.22763, 2026. DOI: 10.48550/arXiv.2605.22763.
- [Verbeken2026] B. Verbeken, B. Vagenende, M.-A. Guerry, A. Algaba, and V. Ginis. Early Evidence of Vibe-Proving with Consumer LLMs: A Case Study on Spectral Region Characterization with ChatGPT-5.2 (Thinking). arXiv:2602.18918, 2026. DOI: 10.48550/arXiv.2602.18918.
- [Woodruff2026] D. P. Woodruff, V. Cohen-Addad, L. Jain, J. Mao, S. Zuo, M.-H. Bateni, S. Branzei, M. P. Brenner, L. Chen, Y. Feng, L. Fortnow, G. Fu, Z. Guan, Z. Hadizadeh, M. T.

Hajiaghayi, et al. Accelerating scientific research with Gemini: Case studies and common techniques. arXiv:2602.03837, 2026. DOI: 10.48550/arXiv.2602.03837.

[Zhang2026] C. Zhang, J. Song, S. Li, Y. Liang, Y. Ma, W. Wang, Y. Zhu, and S.-C. Zhu. Proposing and solving olympiad geometry with guided tree search. *Nature Machine Intelligence* 8 (2026), 84–95. DOI: 10.1038/s42256-025-01164-x.